

A sublacunary interpolation set for distal sequences

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Abstract

Question 6.4 of Le’s paper asks whether no sublacunary set can be an interpolation set for distal sequences. A result already implicit in Pavlov’s 2008 construction and stated explicitly in a 2024 follow-up gives a negative answer. The square set $\{n^2 : n \geq 1\}$ is sublacunary and is an interpolation set for distal systems. We record the two one-line growth computations, identify the precise literature implication, and distinguish this solved question from the source’s two questions that remain open.

1 The exact question

An increasing set $E = \{r_1 < r_2 < \cdots\} \subset \mathbb{N}$ is *sublacunary* in the source if

$$\lim_{n \rightarrow \infty} \frac{\log r_n}{n} = 0.$$

It is an *interpolation set for distal sequences* if every bounded $f : E \rightarrow \mathbb{C}$ extends to a sequence $F(T^n x)$ arising from a continuous function on a compact distal dynamical system. The source asks:

Question 6.4. Is it true that no sublacunary set is an interpolation set for distal sequences?

The exact source page, including the adjacent finite-union question, is reproduced below.

Definition 6.2. A set $E \subset \mathbb{N}$ is called an *interpolation set for distal sequences* if for every bounded function $f : E \rightarrow \mathbb{C}$, there exists a distal sequence $\psi : \mathbb{N} \rightarrow \mathbb{C}$ such that $\psi(n) = f(n)$ for $n \in E$.

Based on Theorem 1.2 and Theorem 1.3, there are two questions we can ask.

Question 6.3. *Let $E \subset \mathbb{N}$ be an interpolation set for distal sequences and $F \subset \mathbb{N}$ finite. Is it true that $E \cup F$ is an interpolation set for distal sequences?*

Question 6.4. *Is it true that no sublacunary set is an interpolation set for distal sequences?*

REFERENCES

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2 The literature criterion

Pavlov’s Corollary 5.1 [2] constructs arbitrary prescribed circle coordinates along sequences whose gaps grow sufficiently rapidly, using iterated skew products on tori. These systems are distal extensions of an irrational rotation. Koutsogiannis, Le, Moreira, Pavlov, and Richter state the exact interpolation consequence as follows [3, Section 7]:

Theorem 1 (Pavlov criterion). *Let $S = \{s_1 < s_2 < \dots\} \subset \mathbb{N}$. If*

$$\limsup_{n \rightarrow \infty} \frac{\log s_{n+1}}{\log(s_{n+1} - s_n)} < \infty, \quad (1)$$

then S is an interpolation set for distal systems.

The original circle-valued realization yields arbitrary bounded complex values by normalization and a product construction. Indeed, every $z = re^{i\theta}$ in the closed unit disk is the average of the two unit complex numbers

$$e^{i(\theta + \arccos r)} \quad \text{and} \quad e^{i(\theta - \arccos r)}.$$

Products of distal systems are distal, and averaging their two continuous coordinate functions gives the required complex extension.

3 The square-set counterexample

Theorem 2. *The set*

$$S = \{n^2 : n \in \mathbb{N}\}$$

is sublacunary and is an interpolation set for distal sequences. Consequently Question 6.4 has a negative answer.

Proof. Put $s_n = n^2$. The criterion (1) applies because

$$\frac{\log s_{n+1}}{\log(s_{n+1} - s_n)} = \frac{2 \log(n+1)}{\log(2n+1)} \rightarrow 2.$$

Thus S is a distal interpolation set. On the other hand,

$$\frac{\log s_n}{n} = \frac{2 \log n}{n} \rightarrow 0,$$

so S is sublacunary in the exact sense of the source. $\square$

The same computation gives a continuum of elementary counterexamples: $s_n = \lfloor n^{1+\varepsilon} \rfloor$ for any $\varepsilon > 0$ (after discarding repetitions, if needed) satisfies (1) and is sublacunary.

4 Status of the adjacent questions

This packet answers only the question for which the literature already gives a complete resolution. The 2024 follow-up explicitly preserves the other two questions:

- The source’s Question 6.3—whether a distal interpolation set remains one after adjoining finitely many integers—reappears as Question 7.1 in [3]. Proposition 7.2 reduces a positive answer to a splitting property for sets of pointwise distal recurrence, and that property remains open.

- The strict hierarchy between k -step and $(k + 1)$ -step interpolation sets is known at $k = 1$ but reappears for $k \geq 2$ as Question 8.2 in [3].

Thus the correct classification is *literature-implied answer*, not a new result: Pavlov’s theorem predates the target, and the later paper by Le and collaborators makes its interpolation consequence explicit.

References

- [1] A. N. Le, *Sublacunary sets and interpolation sets for nilsequences*, Discrete Contin. Dyn. Syst. **42** (2022), 1855–1871; arXiv:2103.13551.
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